

exports to CSV or JSON files can be used for historical analysis.

Google Cloud Platform (GCP) also has many native cloud services for cloud billing; it has reporting, forecasting and optimisation features for compute and storage services, some of which are listed below.

- Cost management provides visualisation reports on service utilisation, filtered by labels. Data Studio can be used to build custom dashboards.
- Billing APIs and data collection APIs (StackDriver, Resource Manager API) offer integration with native and third party cost management tools.
- Google Recommender can be used for optimising cost and usage, based on patterns of resource usage. Resource hierarchy can be used for fine grained resource management for cost allocation.
- Quota limits can be used to proactively control the spend rate on resources including apps and infrastructure.
- Budgets and alerts can throttle cap costs to control spends. These can be used to closely monitor costs, and send alerts through an SMS or e-mail using programmatic budget notifications.
- GCP can be integrated with 'Cloud Build', 'Cloud Pub/Sub' (event/alerts) and cloud functions for cost reporting, billing and notifications.

## Some best practices to handle cloud costs efficiently

Cost management and cost transparency are the two common principles of cloud governance. The FinOps framework addresses both these principles.

The top seven ways to optimise cloud cost usage are listed below.

- Shutdown unused resources/ instances:** A common issue in resource optimisation is handling unused resources or instances, particularly in non-production environments. Monitoring and controlling these will

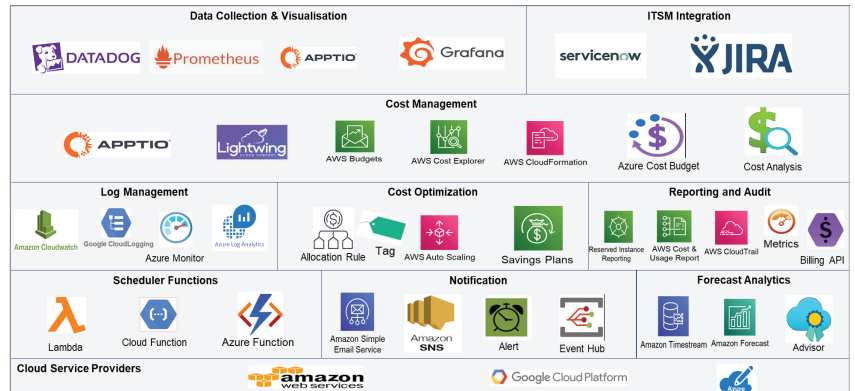


Figure 7: Integrated FinOps toolchain ecosystem

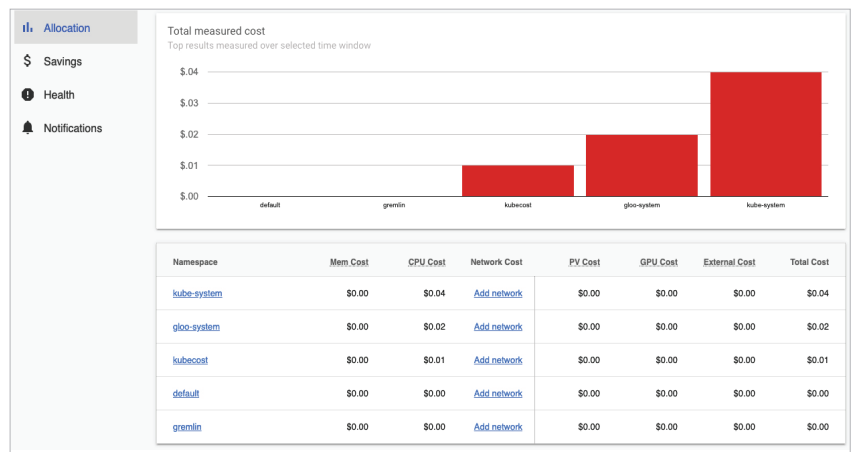


Figure 8: Kubecost dashboard (Image source: Azure documentation)

- enable better cost optimisation.
- Right-size underused resources:** During cloud migration, infrastructure sizing is done based on the existing on-premises infrastructure. We may not know the performance requirements (completely) at the initial stage, and hence right-sizing resources like VM instances, storage services and database size can help reduce costs.
- Reserve instances or spot instances for consistent long-term workloads:** When moving to the cloud, we generally have a clear idea of what we need for long-term usage like data lakes or FTP landing zone. For these services, using reserved or spot instances for three to five years can give as much as 50 per cent cost benefit.
- Choose the hybrid cloud approach for reducing migration costs:**

Instead of moving all applications at once to the cloud, we can migrate in stages, keeping some key applications on-premises.

- Use auto-scaling features for the required resources:** Many cloud services have the auto-scale feature, which helps to scale-up computing and storage when there is higher usage and scale these down in case of lower usage. This feature too helps to reduce costs.
- Budget your resources:** We can set a quota for resources using a budget and cost allocation, so that if there is dynamic burst in resource requirement it can be controlled well within the allocated budget by enabling alerts and notifications.
- Choose the right compute services for better performance and costs:** Though cloud service providers like